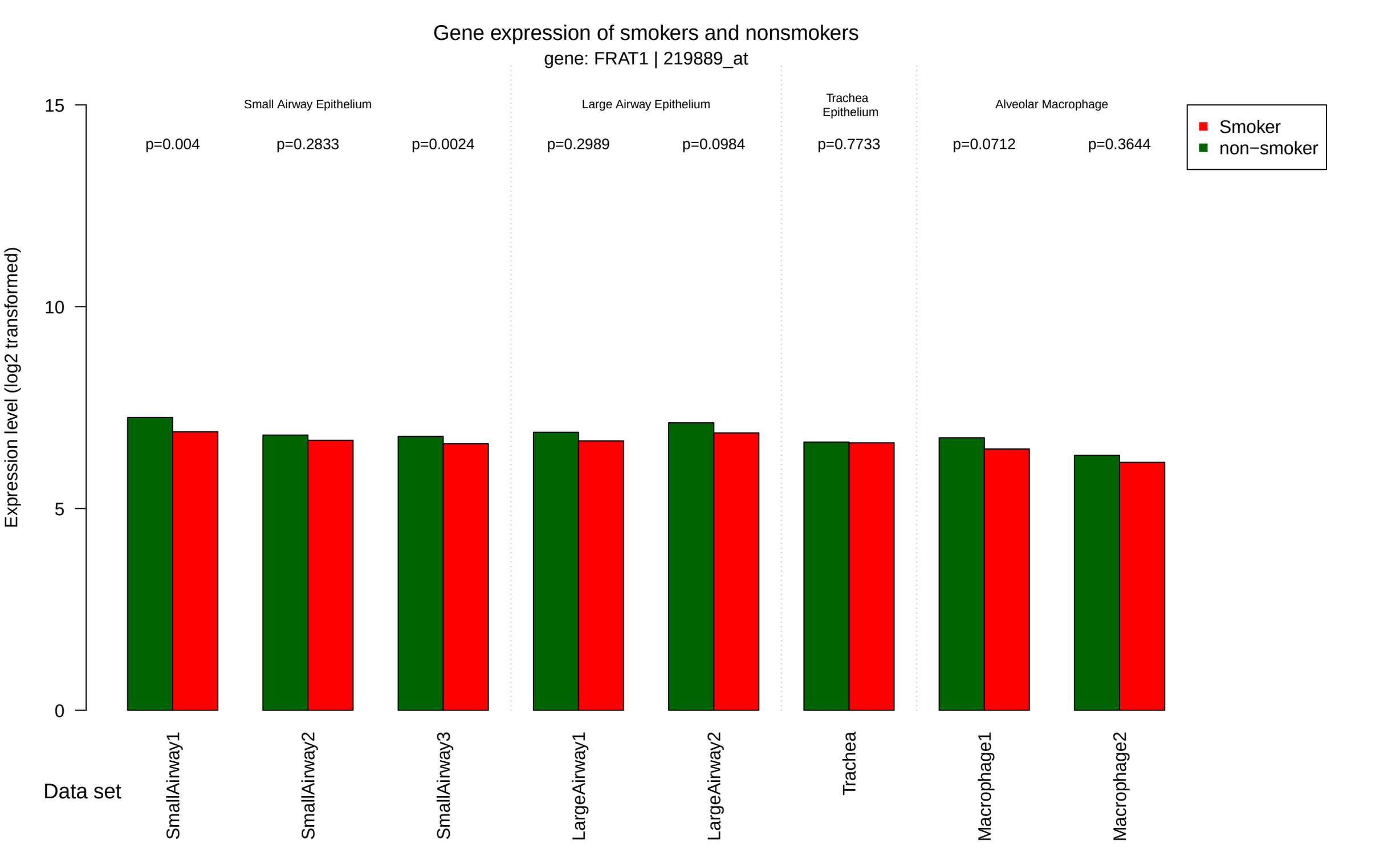
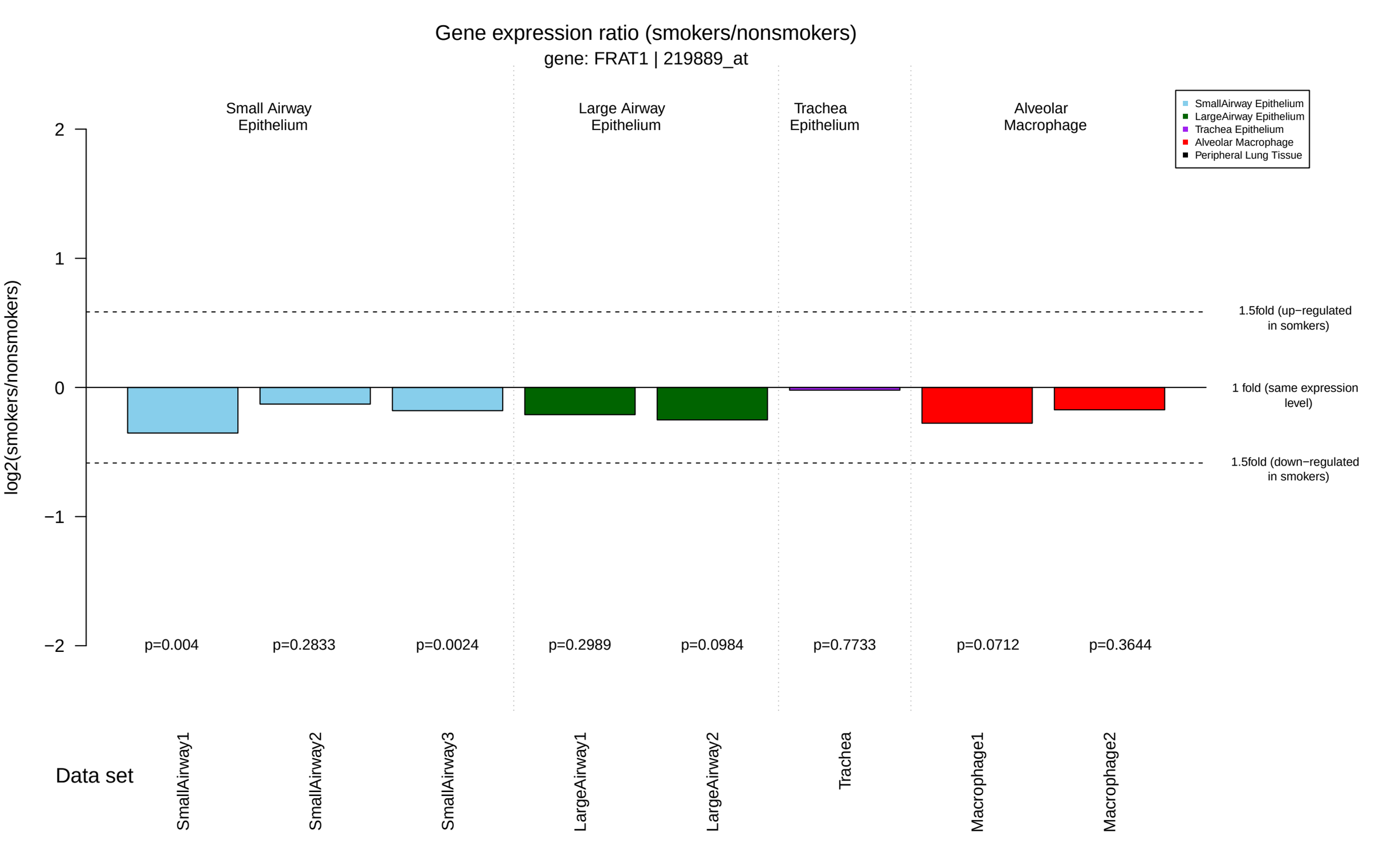
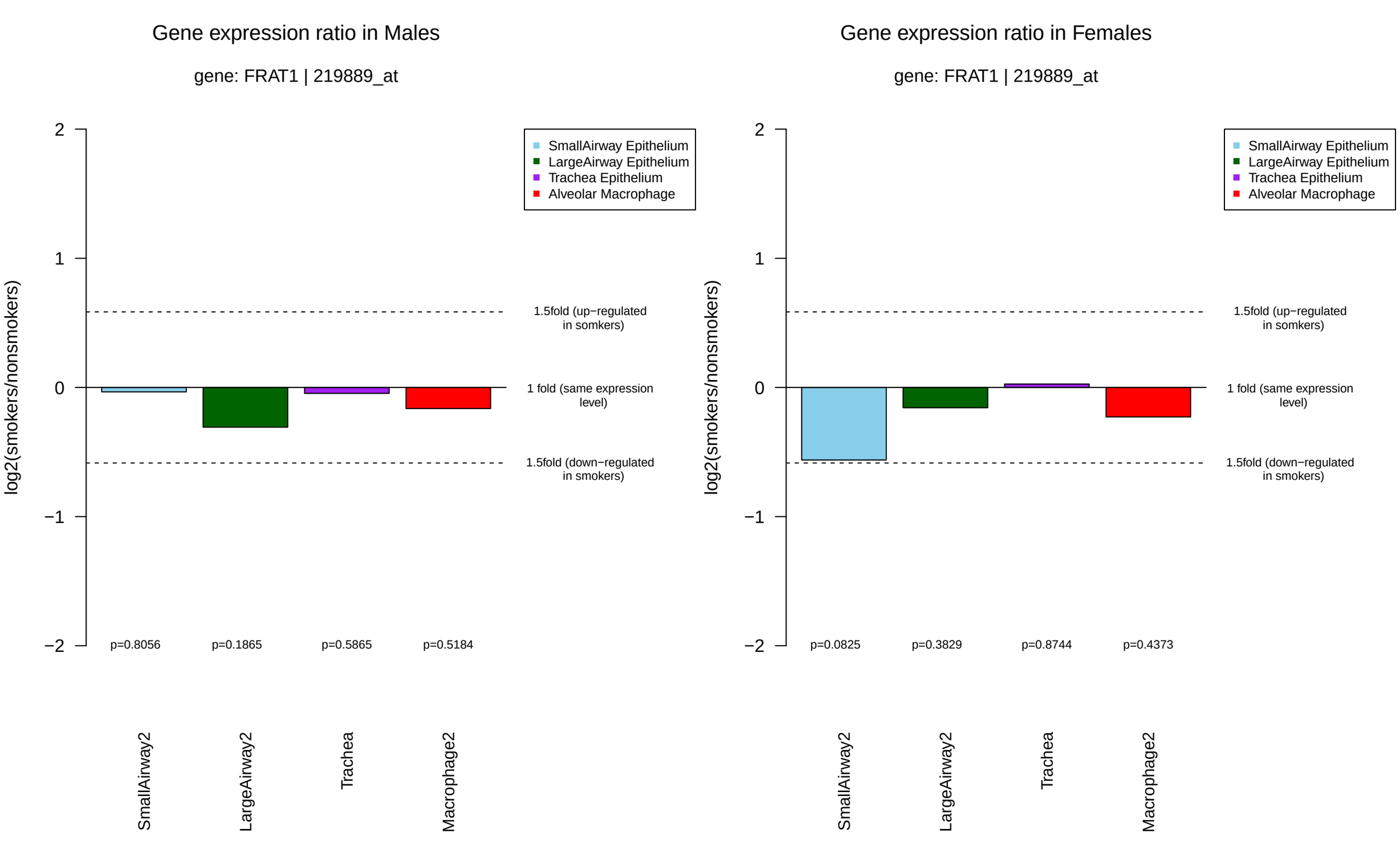
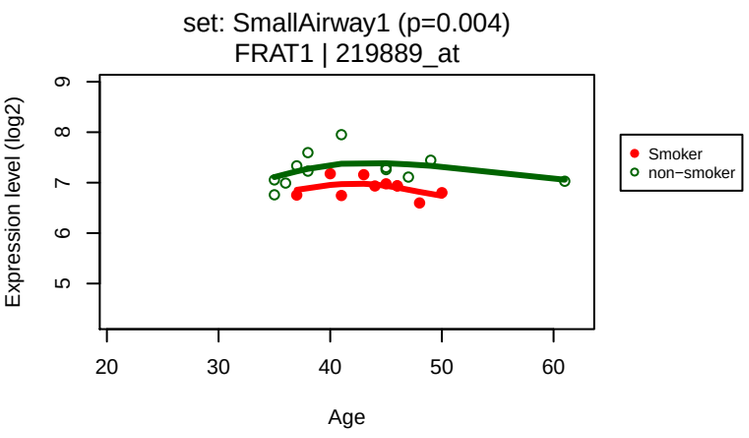
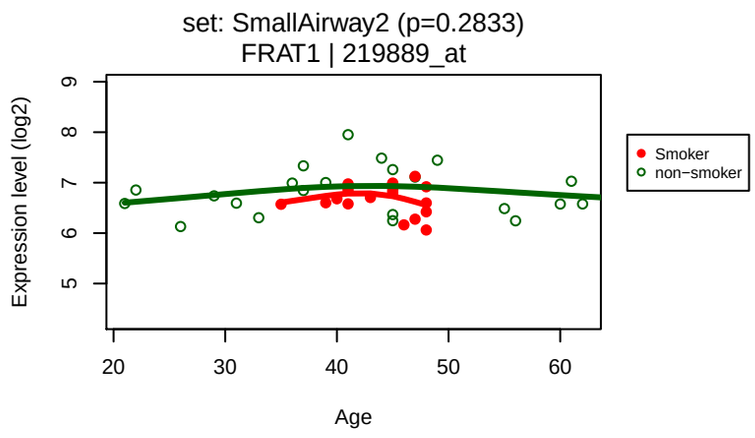
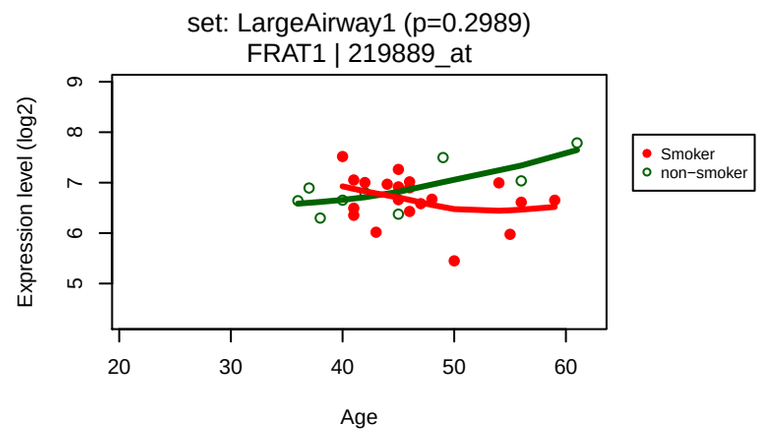

Gene expression of smokers and nonsmokers
gene: FRAT1 | 219889_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.004
p=0.2833
p=0.0024
p=0.2989
p=0.0984
p=0.7733
p=0.0712
p=0.3644
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

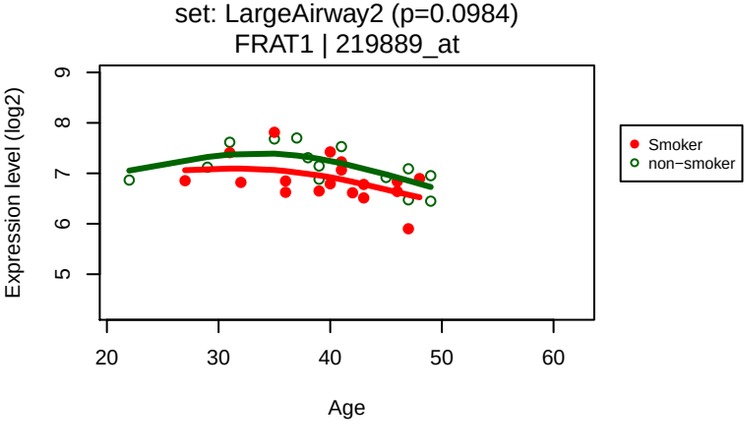

Gene expression ratio (smokers/nonsmokers)
gene: FRAT1 | 219889_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.004
p=0.2833
p=0.0024
p=0.2989
p=0.0984
p=0.7733
p=0.0712
p=0.3644
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

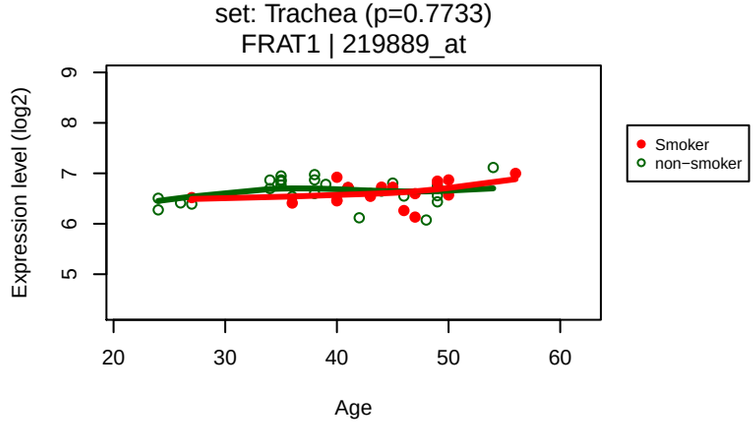

Gene expression ratio in Males
gene: FRAT1 | 219889_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8056
p=0.1865
p=0.5865
p=0.5184
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FRAT1 | 219889_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0825
p=0.3829
p=0.8744
p=0.4373
SmallAirway2
LargeAirway2
Trachea
Macrophage2

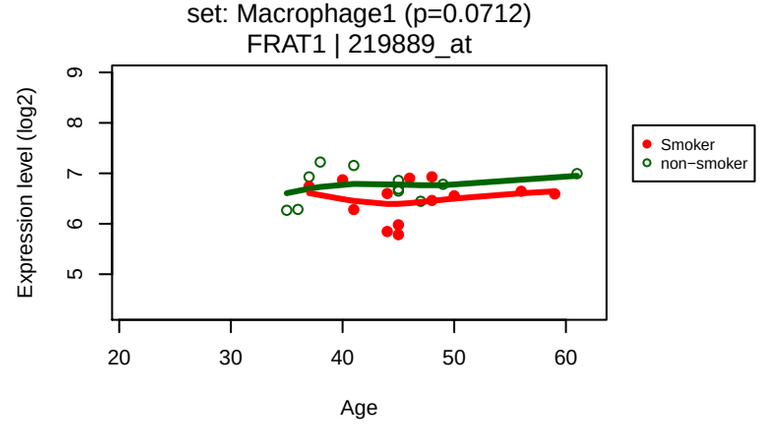

set: SmallAirway1 (p=0.004)
FRAT1 | 219889_at
Expression level (log2)
Age
Smoker
non−smoker

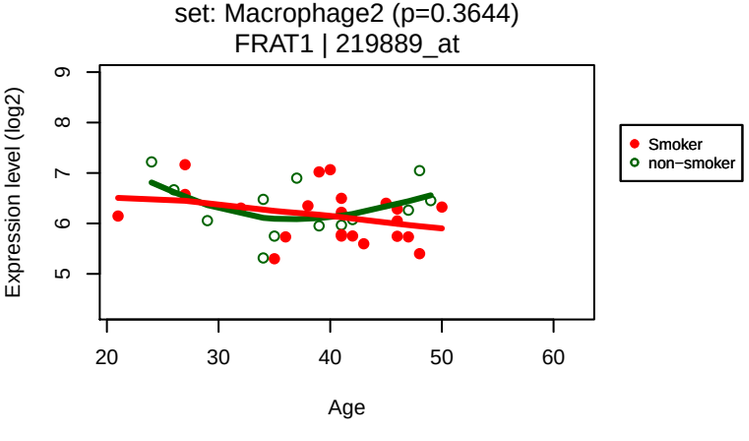

set: SmallAirway2 (p=0.2833)
FRAT1 | 219889_at
Expression level (log2)
Age
Smoker
non−smoker

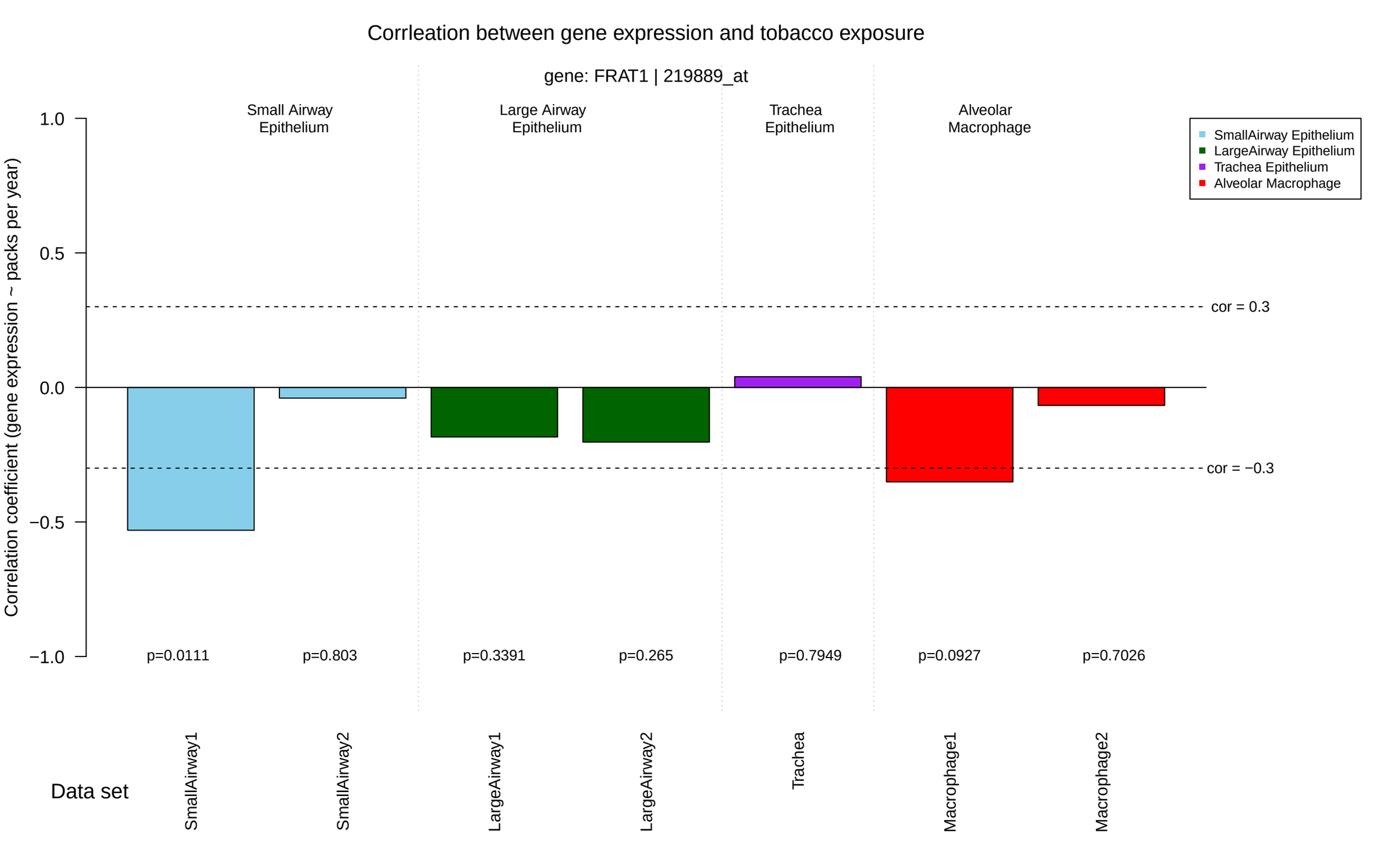

set: LargeAirway1 (p=0.2989)
FRAT1 | 219889_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0984)
FRAT1 | 219889_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7733)
FRAT1 | 219889_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0712)
FRAT1 | 219889_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3644)
FRAT1 | 219889_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FRAT1 | 219889_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0111
p=0.803
p=0.3391
p=0.265
p=0.7949
p=0.0927
p=0.7026
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2